Functional Asymmetry in BERT-Scale Retrieval Encoders: Component Ablation and Implications for Efficient Architecture Design

Karthik Vaddarapu · VIT Vellore · karthikyadav.dev · github.com/zarouz

Problem. The transformer is the default encoder for dense retrieval, yet we lack component-level understanding of *which* sub-modules actually perform retrieval at the BERT scale (110M–335M parameters) where most production systems operate.

Gap. EffiR [1] showed that 30% of layers and most MLP blocks can be pruned from 7B-parameter retrieval encoders with minimal quality loss. Separately, Michalak [2] demonstrated that in hybrid SSM-Transformer architectures, attention is the sole carrier of retrieval signal, with sparsifying to 15% of heads maintaining near-perfect retrieval. Bick et al. [3] showed that preserving just 2% of attention heads (10 heads in a 1B model) recovers over 95% of teacher retrieval performance, with critical heads clustering in upper layers. However, no prior work has performed systematic component ablation on BERT-scale encoders—the architectures that power most production retrieval.

What I did. I ran three inference-time ablation experiments on BGE-base-en-v1.5 (110M params, 12 layers, 144 heads) across three BEIR datasets (SciFact, FiQA, NFCorpus; nDCG@10): (1) progressive back-to-front layer dropping, (2) targeted MLP sublayer removal across layer groups, and (3) importance-ranked attention head pruning (L2 norm of output projection). All experiments are inference-only surgery—no retraining.

Key findings (Fig. 1). *(i) Zero depth redundancy.* Removing a single layer collapses mean nDCG@10 from 0.507 to 0.273 (46% loss); two layers yields 88% collapse—the opposite of 7B-scale behaviour [1]. *(ii) Layer-dependent MLP criticality.* Early-layer MLPs (layers 0–5) are fully load-bearing; removal destroys retrieval (mean $\rightarrow$ 0.004). Upper-layer MLPs (layers 9–11) are nearly expendable: removing all three costs only $\sim$5% quality (0.483 vs. 0.507) with no retraining. *(iii) Upper-layer attention dominance.* At 90% head pruning (15/144 surviving), all survivors cluster in layers 10–11, with every layer-11 head retained. *(iv) Synthesis.* Early layers require both attention and MLPs to build token representations. Late layers primarily perform token-to-CLS information routing, where attention alone suffices. The 5% gap from upper-MLP removal is an *upper bound*—retraining would allow adaptation.

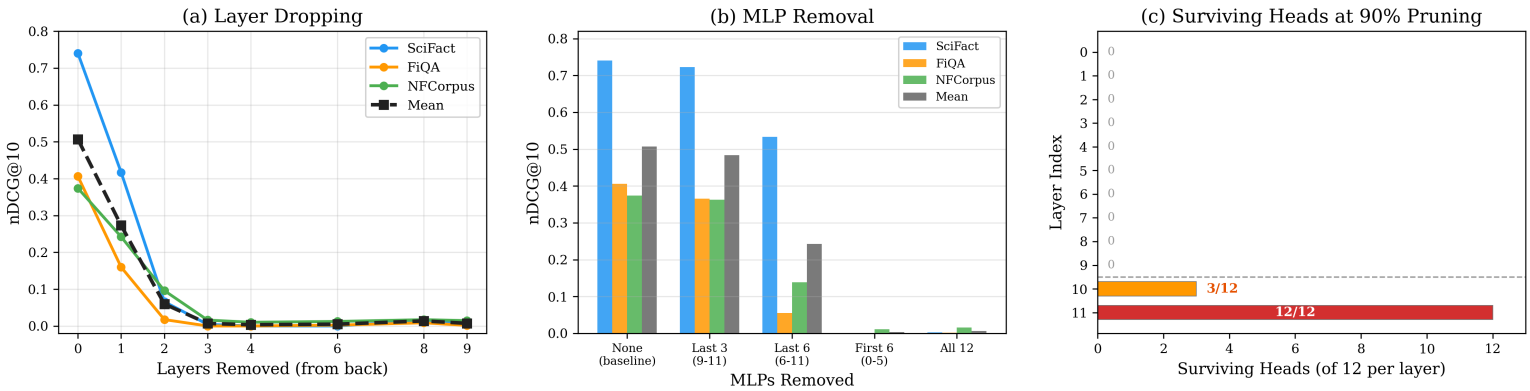

Figure 1: Ablation results on BGE-base-en-v1.5 (110M params, 3 BEIR datasets, nDCG@10).

Proposed 6-month research program. These findings motivate *heterogeneous-block retrieval encoders*: full transformer blocks early, attention-only blocks late. Removing the three upper-layer MLPs already yields 13% parameter reduction (14.2M/109.5M) at 5% quality cost without retraining; distillation should recover this gap. Further directions: (a) whether SSM layers (Mamba [4]) can replace early-layer components via cross-architecture distillation frameworks (MOHAWK/CAB), adapted from language modeling to bi-encoder retrieval; (b) whether upper-layer attention can be further sparsified; (c) full BEIR 13-dataset evaluation against TAS-B, BGE-large-en-v1.5, and E5-Mistral-7B [5].

References

[1] Y. Lei, S. He, A. Li, and A. Yates, "Making large language models efficient dense retrievers," *arXiv:2512.20612*, Dec. 2025.

[2] F. Michalak and S. Abreu, "Some attention is all you need for retrieval," *arXiv:2510.19861*, Oct. 2025.

[3] A. Bick, E. P. Xing, and A. Gu, "Retrieval-aware distillation for Transformer-SSM hybrids," *arXiv:2602.11374*, Feb. 2026.

[4] H. Zhang, C. Chen, L. Mei, Q. Liu and J. Mao, "Mamba Retriever: Utilizing Mamba for Effective and Efficient Dense Retrieval," *arXiv:2408.08066*, Aug. 2024.

[5] L. Wang, N. Yang, X. Huang, B. Jiao, L. Yang, D. Jiang, R. Majumder, and F. Wei, "Text embeddings by weakly-supervised contrastive pre-training," *arXiv:2212.03533*, Dec. 2022.